Government College of Engineering, Chatrapati Sambhajinagar

(An Autonomous Institute of Government of Maharashtra)

F.Y. B. Tech. (All) (NEP) Examination

I S E –II Examination Nov 2023

MABSC 1001: Mathematics-I

Time: 1 Hour *Max. Marks:15* *Date: 08/11/2023* *Time:- 11.30 to 12.30*

"Verify the Course Code and check whether you have got the Correct Question Paper"

N.B.

1. Attempt all questions. 2. Figures to the right indicate full marks 3. Assume suitable data if necessary and state it clearly 4. Use of Non-Programmable Calculator and Data Sheet is allowed

No.	Questions	COs	BT Levels	Marks
1.	**Attempt any two**			
	a) Define gamma function. Evaluate $\int_0^\infty x^7 e^{-2x^2} dx$	CO 1	K-1	**04**
	b) Write trigonometric form of Beta function. Evaluate $\int_0^1 \frac{x^2}{\sqrt{1-x^5}} dx$	CO 2	K-2	**04**
	c) Evaluate $\int_0^\infty 3^{-4x^2} dx$			**04**

P.T.O.

Attempt all questions			
a) Find n'th derivative of $x^3 \sin x$ and hence find 5 th derivative	CO 1	K-1	03
b) State Leibnitz's theorem of successive differentiation If $y = e^{m \sin^{-1} x}$ then prove that $(1-x^2)y_{n+2} - (2n+1)x\,y_{n+1} - (n^2+m^2)y_n = 0$	CO 2	K-2	04
OR			
c) Express $(x-2)^4 - 3(x-2)^3 + 4(x-2)^2 + 5$ in powers of x using Taylor's theorem.			04

Government College of Engineering, Aurangabad

(An Autonomous Institute of Government of Maharashtra)

***F.Y. B.Tech (C/M/CSE/IT)* Examination (NEP)**

End Semester Examination December 2023

MABSC 1001: Mathematics-I

Time: 2Hrs and 30Min. Date of Examination: 18 DEC 2023 *Max. Marks: 60*

"Verify the course code and check whether you have got the correct question paper"

N.B:-

1. *Assume suitable data if necessary and state it clearly*
2. *Use of non-programmable calculator is allowed*
3. *Attempt all questions in sequence*

Q. No.	Questions	Course Outcomes (Cos)	Bloom's Taxonomy Levels (BTLs)	Marks
Q1.	Solve any three a) Expand in polar form $\dfrac{(1+i)^8(\sqrt{3}-i)^4}{(1-i)^4(\sqrt{3}+i)^8}$. b) Find all the roots of $\left(\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)^{\frac{3}{4}}$. c) Prove that $\sin^5\theta=\frac{1}{16}(\sin 5\theta-5\sin 3\theta+10\sin\theta)$. d) Separate real and imaginary parts of $\sin^{-1}(\operatorname{cosec}\theta)$.	CO3, CO5	K2	12
Q2.	Solve any two a) Find nonsingular matrices P *and* Q such that PAQ is in normal form. Also find its rank; where $A=\begin{bmatrix}1&2&-3&-2\\2&-3&1&2\\-1&1&3&2\end{bmatrix}$. b) Solve the following system of equations : $x+y+z=6$ $x+2y+3z=14$ $2x+4y+7z=30$ c) Find eigen values and corresponding eigen vectors for the matrix $A=\begin{bmatrix}3&-1&1\\-1&5&-1\\1&-1&3\end{bmatrix}$.	CO2, CO3, CO5	K2	12

Q3.	Solve any three a) Find n^{th} derivative of $y=\cos^3 x\sin^5 x$. b) If $y=(x^2-1)^n$, prove that $(x^2-1)y_{n+2}+2xy_{n+1}-n(n+1)y_n=0$. c) Expand $\log(\cos x)$ about $\frac{\pi}{3}$. d) Prove that $\sinh^{-1}x=x-\frac{x^3}{6}+\frac{3x^5}{40}-\ldots\ldots\ldots$.	CO2, CO4	K2	12
Q4.	Solve any three a) Evaluate $\int_0^\infty \frac{x^5}{5^x}dx$. b) Evaluate $\int_0^1 \frac{x^2}{\sqrt{1-x^4}}dx.\int_0^1 \frac{dx}{\sqrt{1-x^4}}$. c) Evaluate $\int_0^\infty \frac{e^{-x}}{x}\left[1-e^{-ax}\right]dx, \quad (a>-1)$. d) Show that $erf(\infty)=1$.	CO2, CO4	K2	12
Q5.	Solve any two a) Trace the curve with full justification $xy^2=4a^2(a-x)$. b) Trace the curve with full justification $y^2(a+x)=x^2(b-x)$. c) Trace the curve with full justification $r^2=a^2\cos 2\theta$.	CO5	K3	12

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Government College of Engineering, Aurangabad

(An Autonomous Institute of Government of Maharashtra)

F.Y. (CBCS) Class Test - 1

CSESC1001 : Programming For Problem Solving

Time: One Hour Date of Examination: 05/10/2023 *Max. Marks: 15*

erify the course code and check whether you have got the correct question paper"

N.B:-

1. *Assume suitable data if necessary and state it clearly*
2. *Use of non-programmable calculator is allowed*
3. ***Attempt Any Three .***

Q.No.	*Question*	Course Outcomes (Cos)	Bloom's Taxonomy Levels (BTLs)	Marks
Q1	Write an algorithm and draw flowchart for following C programs: 1) Program to accept 5 subject marks and print its total and average. 2) Program to find largest number among three numbers.	CO1	K1	05
Q2	Summarize any two of the following computer components: I. Memory ii. Processor iii. I/O devices	CO1	K1	05
Q3	Describe Operators in C.	CO1	K1	05
Q4	Write a C program to check whether a given number is even or odd.	CO2	K2	05

Government College of Engineering, Chhatrapati Sambhaji Nagar

(An Autonomous Institute of Government of Maharashtra)

FYBTECH – NEP – SEMESTER – I

End Semester Examination December 2023

CSESC1001:Programing for problem solving

Time: 2 Hours and 30 Minutes 15 DEC 2023 ***Max. Marks: 60***

"Verify the Course Code and check whether you have got the Correct Question Paper"

N.B.

1. ***Attempt all questions.***
2. ***Figures to the right indicate full marks.***
3. ***Assume suitable data if necessary and state it clearly.***
4. ***Use of Non-Programmable Calculator and Data Sheet is allowed.***

Q1	**Attempt any two**	***CO***	***B.T. Level***	***Marks***
A	Describe architecture of a computer system in detail	*1*	*1*	*06*
B	Define algorithms and flowcharts in detail. Write an algorithm and flowchart for adding two numbers.	*2*	*2*	*06*
C	Explain standard Input output in C.	*2*	*1,2*	*06*
Q2	**Attempt any two**	***CO***	***B.T. Level***	***Marks***
A	Summarize the different data types in C with examples.	*2*	*2*	*06*
B	Enlist and explain the following operators with example: 1. Arithmetic operators 2. Logical operators 3. Relational operators	*2*	*2*	*06*
C	Write a program to accept a character and convert into an integer and display the value.	*3*	*2*	*06*
Q3	**Attempt any two**	***CO***	***B.T. Level***	***Marks***
A	Describe syntax of the following with examples: 1. if statement 2. if else 3. if else ladder	*2*	*2*	*06*
B	Write a program to input sides if a triangle and check whether it is equilateral(All sides equal), isoceles(Two sides equal), scalene(No side equal)	*3*	*3*	*06*
C	Describe the use and syntax of Switch case. Write a program to input a single digit number and convert it into it's equivalent word. Eg: Input: 1 Output: One	*2,3*	*2,4*	*06*
Q4	**Attempt All**	***CO***	***B.T. Level***	***Marks***
A	Summarize the use and syntax of 1-Dimensional arrays. Write a program to declare and initialize and display contents of 1-D array	*3*	*2,3*	*06*
B	Write a short note on using Strings in C. Explain with the help of a program	*2,3,5*	*2,3*	*06*
Q5	**Attempt All**	***CO***	***B.T. Level***	***Marks***
A	Differentiate between union and structures	*4*	*3,4*	*06*
B	Explain the concept of pointers with a program.	*4,5*	*3*	*06*

Note: CO: Course Outcome; B.T. Level: Bloom's Taxonomy Level

Government College of Engineering, Aurangabad

(An Autonomous Institute of Government of Maharashtra)

FY.B.Tech ISE –I Examination OCT- 20223

CHBSC: 1001 (ELECTROCHEMISTRY, BATTERY SCIENCE AND ENGINEERING MATERIA

Time: 01 Hr. *Max. Marks: 15*

Q1	**Attempt any TWO of the following**	***CO***	***B.T. Level***	***Marks***
A.	Describe construction and working of Li –ion battery with electrode reactions and mention their applications?	***CO3***	***K3***	***04***
B.	With neat labelled diagram explain construction and working of bomb calorimeter and derive equation for determination of GCV with necessary corrections?	***CO2***	***K2***	***04***
C	Calculate GCV and NCV of a fuel from the following data. Mass of fuel=0.75gm, w=350gm ,Rise in temp=3.02 C, Mass of water = 1150gm, % H2=2.8, Latent heat of steam=587 KJ/kg	***CO4***	***K4***	***04***
Q2	**solve the followings(All questions compulsory)**	***CO***	***B.T. Level***	***Marks***
A.	Derive the Nernst equation for electrode potential.	***CO5***	***K5***	***04***
B	Define battery. Give classification of battery With examples.	***CO3***	***K3***	***03***

GOVTRNMENT COLLEGE OF ENGINEERING, CHHATRAPATI SAMBHAJINAGAR

(An Autonomous Institute of Government of Maharashtra)

FY.B.Tech ISE –II Examination Nov- 2023

CHBSC1001: ELECTROCHEMISTRY, BATTERY SCIENCE AND ENGINEERING MATERIALS

ne: 01 Hr. *Max. Marks: 15*

Q1	**Attempt any TWO of the following**	*CO*	*B.T. Level*	*Marks*
A.	Define conducting polymers? Explain conduction mechanism of polyacetylene with suitable figures?	*CO4*	*K4*	*04*
B.	What are LCP? Describe Nematic and smectic liquid crystals with figure and give their application?	*CO3*	*K3*	*04*
C	Explain properties and applications of biodegradable polymers by taking examples of Polylactic acid	*CO3*	*K3*	*04*
Q2	**SOLVE THE FOLLOWING (all questions compulsory)**	*CO*	*B.T. Level*	*Marks*
A.	Write down tweleve principals of green chemistry and explain any two with examples?	*CO2*	*K2*	*04*
B	Define MAOS? Give merities and demerities of MAOS with any two examples?	*CO4*	*K4*	*03*

F.Y B.TECH (NEP) Examination SEM -1

End Semester Examination December 2023

20 DEC 2023

e: 2 Hours 30 Minutes Max. Marks: 60

"Verify the Course Code and check whetheryou have got the Correct Question Paper"

ttempt all questions.
igures to the right indicate full marks
ssume suitable data if necessary and state it clearly
se of Non-Programmable Calculator and Data Sheet is allowed

Q1	**Attempt any TWO**	CO	B.T. Level	Marks
a	Define Nano materials and how are they classified based on the dimension? Sketch the CVD method for preparation of Nanomaterials?	CO3	BT3	06
b	Explain the following i) Reserve Battery. ii) Power density and Energy Density III.Shelf life	CO3	BT3	06
c	During the determination of calorific value of a gaseous fuel by Bomb calorimeter, the following results were recorded (i) Weight of Fuel Burnt = 1.90 gm. (ii) Water Equivalent of Calorimeter = 500 gm. (iii) Weight of water taken in calorimeter = 2300 gm(iv) Initial Temperature of water=25.05° C. (v) Final Temperature of water = 29.23°C. vi) Acid Correction=15 cal(vii) Fuse wire Correction = 7cal .. (Viii) Cooling Correction=0.26°C . If the fuel contains7.5% Hydrogen, Calculate the gross and net calorific value of coal sample provided that the latent of steam condensed is 587 cal/gm .	CO4	BT4	06
Q2	**Attempt any TWO**	CO	B.T. Level	Marks
a.	What are LCP? Explain Nematic and sematic type in detail With figures?	CO2	BT2	06
b	Mention the advantages of MAOS With suitable examples? What are the criteria for selecting a solvent for microwave assisted reactions?	CO3	BT3	06
c	Define Battery? give characteristics of battery and explain, basic requirements for commercial batteries? .	CO2	BT2	06

Q3	**Attempt any TWO**	*CO*	*B Lev*
a.	A sample of coal contains C = 93%, H = 6%, and Ash=1%. The following data were obtained when the above coal was tested in Bomb calorimeter. 1) Weight of coal burned = 0.92gm 2) Weight of water taken = 2200gm 3) water equivalent of Bomb calorimeter = 550gm 4) rise in temperature = 2.42oC 5) Fuse wire correction = 10cal 6) Acid correction= 50cal. Calculate the Gross and Net calorific value of the coal sample. (Latent heat of condensation of steam = 580cal/kg	*CO4*	*BT 4*
b	Differentiate between Battery and fuel cells	*CO4*	*BT4*
c	Explain in brief carbon Nano tubes. What are the various types of carbon Nano tubes? Discuss the Synthesis and properties of carbon Nano tubes?	*CO2*	*BT2*
Q4	**Attempt any TWO**	*CO*	*B.T. Level*
a	Explain the construction and working of Li ion batteries with its advantages, limitation and application. ?	*CO4*	*BT4*
b	What are conducting polymers? State their properties and application with examples?	*CO2*	*BT2*
c	Write short notes on i N-doping ii P- doping	*CO3*	*BT3*
Q5	**Attempt any TWO**	*CO*	*B.T. Level*
a	Explain Fullerenes and give their properties and applications?	*CO2*	*BT2*
b	With net cell reaction explain construction and working of solid oxide fuel cell and their applications?	*CO3*	*BT3*
C	State and explain 12 principles of green chemistry and explain supercritical fluids and ionic liquids?	*CO2*	*BT2*

Note: CO: Course Outcome; B.T. Level: Bloom's Taxonomy Level

(An Autonomous Institute of Government of Maharashtra)

F.Y. Examination (NEP)

(CSE and IT)

Class Test – II

CEESC 1001: BASICS OF CIVIL ENGINEERING

Date of Examination: 28th November 2023

mins

"Verify the course code and check whether you have got the correct question paper"

	Course Outcomes (CO)	Bloo Taxo Le
ttempt any two:		
Write a short note on remote sensing.	CO2	
List down the objectives of surveying.	CO2	
Convert WCB to RB and RB to WCB: 1. $56^0 30'$ 2. 136^0 3. $328^0 30'$ 4. S 46^0 E 5. N 35^0W	CO2	
Following consecutive observations were taken with a dumpy level: 2.515, 2.325, 2.195, 1.710, 1.525. Calculate RL of all the points if RL of BM is 500m. Also give the check.	CO2	

Government College of Engineering, Aurangabad
(An Autonomous Institute of Government of Maharashtra)
F.Y. Examination (NEP)
(CSE and IT)
Class Test – I
CEESC 1001: BASICS OF CIVIL ENGINEERING

Time: 45 mins | **Date of Examination: 28th November 2023** | Max. Marks: 10

"Verify the course code and check whether you have got the correct question paper"

	Course Outcomes (CO)	Bloom's Taxonomy Levels	Marks
Q1. Attempt any two:			
i. List down all the principles of planning and explain any two in detail.	CO1	1	5
ii. Differentiate between load bearing structure and framed structure.	CO1	4	5
iii. Define: 1. Covered area 2. Built up area 3. Plot 4. Carpet area 5. FSI	CO1	1	
iv. List down the points which should be taken into consideration while selecting a site for a building.	CO1	1	

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F.Y.B.TECH. Civil (NEP) Examination

End Semester Examination Nov-Dec 2023

CEESC 1001 : Basics Of Civil Engineering

Time: 1 Hour 30 mins 13 DEC 2023 **Max. Marks: 30**

"Verify the Course Code and check whether you have got the Correct Question Paper"

N.B.

1. ***Attempt all questions.***
2. ***Figures to the right indicate full marks.***
3. ***Assume suitable data if necessary and state it clearly.***
4. ***Use of Non-Programmable Calculator and Data Sheet is allowed.***

Q1	**Attempt any TWO**	***CO***	**B.T. Level**	***Marks***
A	Which points should be considered in the selection of site for: 1. Residential Building 2. Commercial Building	1	2	***05***
B	What are the building Bye- laws? And state the necessities of building Bye laws.	1	4	***05***
C	What is the importance of "Economy" in planning of a building?	1	4	***05***
Q2	**Attempt any TWO**	***CO***	**B.T. Level**	***Marks***
A	Write in detail about remote sensing.	2	2	***05***
B	Describe in brief the temporary adjustments Dumpy level.	2	2	***05***
C	Convert WCB to RB : 1. 136^0 - 44 2. $328^0 30'$ 3. $56^0 30'$ 4. $238^0 30'$ 5. 325^0	2	4	***05***
Q3	**Attempt any TWO**	***CO***	**B.T. Level**	***Marks***
A	What are the properties of a good building stone?	3	4	***05***
B	Explain various equipments used for excavation.	4	2	***05***
C	Brief on watershed management.	5	3	***05***

Government College of Engineering, Aurangabad
(An Autonomous Institute of Government of Maharashtra)
***B.E. Civil* Examination**
Class Test – I
***CS IKS 1003*: Indian Knowledge System**

Time: One Hour Date of Examination: -10-2023 *Max. Marks: 10*

"Verify the course code and check whether you have got the correct question paper"

N.B:-

1. *Assume suitable data if necessary and state it clearly*
2. *Use of non-programmable calculator is allowed*
3. *Attempt all questions in sequence*

		Course Outcomes (Cos)	Bloom's Taxonomy Levels (BTLs)
Q1.	**Multiple choice questions**		
i.	The theory of reason is also known as _____. a. Atma vidya b. Hetu Shastra c. Param Vidya d. Aparam Sastra	CO2, CO3, C04, CO5	K2
ii.	When the knowledge is gained through comparision it is known as_____. a. Pratyaksha b. Upamana c. Anumaana d. Sabda		
Q2.	**Attempt any one**	CO1, CO2, CO3, CO4, CO5	K2
i.	Explain the various types of Pramana (atleast 4) and describe the Vedas in short.		
ii.	Explain the triangle of knowledge and draw the flowchart of knowledge (Buddhi).		
Q3.	**Attempt any one**	CO2, CO3, C04, CO5	K2
iii.	What is logic and explain the logical fallacies?		
iv.	Explain in Indian knowledge system in detail and the vedic corpus i.e. Vedas and Vedangas		

(An Autonomous Institute of Government of Maharashtra)

F.E. CSE Examination

Class Test – II

ET IKS 1001 : Indian Knowledge System

Time: 45min Date of Examination: 06 -11-2023 Max. Marks: 10

"Verify the course code and check whether you have got the correct question paper"

N.B:-

3. *Assume suitable data if necessary and state it clearly*
4. *Use of non-programmable calculator is allowed*

Q1. Multiple choice questions

i.	_____was the master of Prosody. b. Pingalacharya b. Mahavira c. Bramhagupta d. Aryabhata	CO2, CO3,	K2
vi.	Unit to measure utterance in a syllable is known as_____. b. Vritta b. Akshara c. Mora d. Meru	C04, CO5	

Q2. Attempt any one

iii.	Explain the unique aspects of Indian Mathematics	CO1, CO2,	K2
iv.	Explain the historical development of astronomy in India	CO3	

Q3. Attempt any one

i.	Describe the discovery of zero and its importance	CO2,	K2
i.	Explain the Indian calendar of Panchanga	C04, CO5	

Government College of Engineering, Chhatrapati Sambhajinagar
(An Autonomous Institute of Government of Maharashtra)
End Semester Examination 2023
First Year B.Tech. C.S.E. Examination
CSIKS-1003: Indian Knowledge System

Time: 1.30 Hour. Max. Marks: 30

"Verify the Course Code and check whether you have got the Correct Question Paper"

N.B.

11 DEC 2023

1. Attempt all questions.
2. Figures to the right indicate full marks
3. Assume suitable data if necessary and state it clearly
4. Use of Non-Programmable Calculator and Data Sheet is allowed

Q1	Attempt any two	CO	Level	Marks
	a. Discuss how Sanskrit can be used for natural language processing by Panini.	1	K-1	05
	b. Explain the triangle of knowledge and draw the flowchart of knowledge.	2	K-1	05
	c. Explain the various pramanas and illustrate Vedas in detail.	1	K-1	05
Q2	Attempt any two	CO	Level	Marks
	a. What is IKS, explain in detail about the vedic corpus?	3	K-1	05
	b. Explain the elements of Indian Calendar.	3	K-1	05
	c. Describe the characteristics of the Indian number system and its importance in science and technology.	3	K-1	05
Q3	Attempt any two			
	a. Explain the Historical development of celestial coordinate system India and the related astronomical system.	4	K-2	05
	b. Describe the unique aspects of Indian maths typically the discovery of 0.	3	K-2	05
	c. What is logic and explain the logical fallacies?	4	K-2	05

Government College of Engineering, Aurangabad

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F.Y.(CBCS) Class Test - 1

CSPCC1004/ITPCC1001- COMPUTER ORGANIZATION

Time: 40min Date of Examination: 04/04/2024 *Max. Marks: 10*

"Verify the course code and check whether you have got the correct question paper" *N.B:-*

1. *Assume suitable data if necessary and state it clearly*
2. *Use of non-programmable calculator is allowed*
3. ***Attempt Any TWO .***

Q.No.	***Question***	Course Outcomes (Cos)	Bloom's Taxonomy Levels (BTLs)	Marks
Q1	Difference between Saas, Paas, Iaas.	CO1	K3	05
Q2	Explain evolution of the Intel x86 Architecture?	CO1	K3	05
Q3	Describe Interconnection structure.	CO1	K4	05

Government College of Engineering, Aurangabad

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F.Y. (CBCS) Class Test - 2

CSPCC1004/ITPCC1001- COMPUTER ORGANIZATION

Time:40 m Date of Examination: 28/03/2024 *Max. Marks: 10*

"Verify the course code and check whether you have got the correct question paper" *N.B:-*

1. *Assume suitable data if necessary and state it clearly*
2. *Use of non-programmable calculator is allowed*
3. ***Attempt Any TWO .***

Q.No.	*Question*	Course Outcomes (Cos)	Bloom's Taxonomy Levels (BTLs)	Marks
Q1	Draw and explain Solid state drive Architecture?	CO2	K3	05
Q2	What is DMA ? Explain block diagram of DMA. Also describe how DMA is used to transfer data from peripherals.	CO2	K4	05
Q3	Write short note on Interrupt driven I/O?	CO2	K3	05

Government College of Engineering, Chhatrapati Sambhajinagar

(An Autonomous Institute of Government of Maharashtra)

B.Tech. First Year In Semester Examination I

INAEC1001: Communication skills

Date: 13th April 2024 ***Time: - 45Min*** ***Max. Marks: 10***

Q.No. 1	**Solve any one**	***CO***	***B.T. Level***	***Marks***
(A)	What is parts of speech? Explain any five with examples	*2*	*K3*	*05*
(B)	What is communication? Explain the concept			
Q.No. 2	**Define the terms**			
(A)	Upward communication	*1*	*K2*	*03*
(B)	Informal communication			
(C)	Internal Communication			
Q.No. 3	**Write the Formulas**			
(A)	Perfect Future Tense	*2*	*K1*	*02*
(B)	Continuous Past Tense			

Note: CO: Course Outcome; B.T. Level: Bloom's Taxonomy Level

Government College of Engineering, Chhatrapati Sambhajinagar

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B.Tech. First Year In Semester Examination II

INAEC1001: Communication skills

Date: 27th March2024 ***Time: - 45Min*** ***Max. Marks: 10***

Q.No. 1	**Solve any one**	*CO*	*B.T. Level*	*Marks*
(A)	Write a report on tree plantation programme organised by your college	*2*	*K3*	*05*
(B)	Write a memo to your junior clerk for his absence on duty without prior permission.			
Q.No. 2	**Explain following with one example each**			
(A)	Posture	*1*	*K2*	*03*
(B)	Silence			
(C)	Dress & Appearance			
Q.No. 3	**Fill in the blanks**			
(A)	Nonverbal communication concerned with such as.....(sound, wood)	*2*	*K1*	*02*
(B)	Every verbal communication requires ..... codes for effective communication (verbal, nonverbal)			

Note: CO: Course Outcome; B.T. Level: Bloom's Taxonomy Level

(An Autonomous Institute of Government of Maharashtra)

...gineering, Chh. Sambhajinagar

B.Tech. First Year

End Semester Examination April/ May 2024

INAEC1001: Communication skills

Time: 1:30 Hours *Max. Marks: 30*

"Verify the Course Code and check whether you have got the Correct Question Paper"

06 MAY 2024

N.B.

1. *Attempt all questions.*
2. *Figures to the right indicate full marks*
3. *Assume suitable data if necessary and state it clearly*

Q. 1	**Attempt Any Two of the Following**	*CO*	*B.T. Level*	*Marks*
(A)	Draft a notice informing students about a sport meet, giving schedule of activities	*2*	*K2*	*05*
(B)	Explain the process of communication with the help of its different element.	*2*	*K2*	*05*
(C)	As General secretary, student association, draft circular informing the students about the book bank facility made available in their town.	2	K2	05
Q2	**Attempt Any Two of the Following**			
(A)	What is the psychological barrier? Explain any four with remedies.	*3*	*K3*	*05*
(B)	What is Email communication? state the importance of Email – Etiquette in the workplace	*3*	*K3*	*05*
(C)	What is nonverbal communication? Explain different aspects of body language	*3*	*K3*	05
Q3	**Solve the following as directed.**			
(A)	**Write short note on**			
1	Voice Modulation	*1*	*K1*	*02*
2	Professional clothing etiquette	*1*	*K1*	*02*
(B)	**Solve the following**			
1	Write the synonyms of following i) Reel ii) Upright	*1*	*K1*	*02*
2	Write the antonyms of following i) Talent ii) Kindle → Dumb	*1*	*K1*	*02*
3	Identify the Adverb	1	*K1*	02
	i) The post man comes to her daily.			
	ii) The mangoes are almost ripe.			

Note: CO: Course Outcome; B.T. Level: Bloom's Taxonomy Level;

Government College of Engineering Aurangabad, Chhtrapati Sambhajinagar

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F. Y. B. Tech. (All Branches) Examination (NEP)

ISE–I Feb 2024

MABSC 1003: Mathematics-II (Civil, Mech, CSE, IT)

MABSC 1004: Mathematics-II (Electrical, E&TC)

Time: 03.20 to 04.20 Date of Examination: 16.02.2024 Max. Marks: 15

"Verify the course code and check whether you have got the correct question paper"

N.B:-1.Assume suitable data if necessary and state it clearly 2.Use of non-programmable calculator is allowed 3. Attempt all questions in sequence

Q. No.	Questions	COs	BT Levels	Marks
1.	Attempt any two sub questions			
	a) Define Exact Differential Equation. Evaluate $\left(y^2.e^{xy^2}+4x^3\right)dx+\left(2xy.e^{xy^2}-3y^2\right)dy=0$	CO 1, CO3,	K -1 K-2	04
	b) Define Bernoulli's Differential Equation. Evaluate $\cos y - x\sin y\frac{dy}{dx}=\sec^2 x$	CO4		04

Que 1.	c) A pipe 18 cm in diameter contains steam at $160^{0}C$. It is covered with asbestos, 6 cm thick, for which $k = 0.0006$ and the outer surface is at $50^{0}C$. Find the amount of heat lost per hour from a 12 meter long pipe.	CO 5	K-2
Que 2.	Attempt All Sub questions		
	a) If $u = \sin^{-1}\left(\frac{x^{1/6} + y^{1/6}}{x^{1/9} - y^{1/9}}\right)$, find $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2}$.	CO 3	
	b) If $f\left(x+y+z,\ x^2+y^2+z^2\right) = 0$, Prove that $(y-x)+(y-z)\frac{\partial z}{\partial x}+(z-x)\frac{\partial z}{\partial y} = 0$.	CO 2	K-2
	OR		
	c) If $u = x+y+z$, $v = x-y+z$, $w = x^2+y^2+z^2+2xz$ Examine whether u, v, w are functionally dependent. If so find the relation between them.	CO 4	

Government Col

...gabad, Chhatrapati Sambhajinagar
...Institute of Government of Maharashtra)
...T. B.Tech (Civil/Mech/CSE/IT) Examination (NEP)
End Semester Examination May-June 2024

Time: 2 Hours 30Mins

MABSC 1003 : Mathematics II

Date of Examination: 09 MAY 2024

Max. Marks: 60

"Verify the course code and check whether you have got the correct question paper"

N.B:-

1. *Assume suitable data if necessary and state it clearly*
2. *Use of non-programmable calculator is allowed*
3. *Attempt all questions in sequence*

Q. No.	Questions	Course Outcomes (Cos)	Bloom's Taxonomy Levels (BTLs)	Marks
Q1.	Solve any four			
	a) Solve $\frac{dy}{dx}+\frac{y\cos x+\sin y+y}{\sin x+x\cos y+x}=0.$	CO 1	K1	12
	b) Find the Jacobian of the function $u=x^2-y^2,\ v=2xy.$			
	c) Find $grad\ \phi$ at $(2,-2,-1)$ if $\phi=2xz^4-x^2y.$			
	d) Find the Fourier series of $f(x)=\frac{\pi+x}{2}$ in the interval $(0,2\pi).$			
	e) Evaluate $\int_0^1\int_0^{\sqrt{1+x^2}}\frac{dxdy}{1+x^2+y^2}.$			
Q2.	Solve any three	CO2	K2	
	a) Solve the differential equation $\frac{dy}{dx}+y\cot x=\cos x.$			
	b) If $y\log(\cos x)=x\log(\sin y)$, find $\frac{dy}{dx}.$			
	c) Evaluate $\iint e^{3x+4y}dxdy$, over the triangle bounded by the lines $x=0, y=0$ and $x+y=2.$			
	d) Find the half range sine series of $f(x)=x-x^2$ in $0<x<1.$			
Q3.	Solve any three	CO3	K2	
	a) Solve the differential equation $\frac{dy}{dx}=\frac{x^2-2xy}{x^2-\sin y}.$			
	b) If $u=\tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, prove that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\sin 2u.$			
	c) Show that $\bar{F}=(y^2-z^2+3yz-2x)\hat{i}+(3xz+2xy)\hat{j}+(3xy-2xz+2z)\hat{k}$ is both solenoidal and irrotational.			

	d) Find the Fourier series of the function $f(x)=\frac{x(\pi^2-x^2)}{12}$ in the interval $(-\pi,\pi)$.			
Q4.	Solve any three a) Solve the differential equation $ydx+x(1-3x^2y^2)dy=0$. b) Find maxima and minima of the function $f(x,y)=x^2+y^2+6x-12$. c) Change the order of the integration and evaluate $\int_0^1\int_x^{2-x}\frac{x}{y}\,dx\,dy$. d) Find the work done in moving a particle from $A(1,0,1)$ *to* $B(2,1,2)$ along the straight line AB in the force field $\vec{F}=x^2\hat{i}+(x-y)\hat{j}+(y+z)\hat{k}$.	CO4	K2	12
Q5.	Solve any two a) A voltage of $20\cos 5t$ is applied to a series circuit consisting of 10 *ohm* resistor and 2 *henry* inductor. Find the current at any time t. b) Find the percentage error in calculating the area of a rectangle when an error of 2 % is made in measuring each of its side. c) Find the area common to the circles $x^2+y^2-4y=0$ and $x^2+y^2-4x-4y+4=0$.	CO5	K3	12

Government College of Engineering, Chhatrapati Sambhajinagar

B. Tech. First Year **Re-In Semester Examination 1**

PHBSC1002: Optics, Semiconductors & Quantum Mechanics

Date: April 19, 2024 Time: 11.00 to 12.00 Max. Marks: 15

Q. N.	**Solve any three.**	CO	B.T. Level	Marks
1	What is Fermi level? Discuss Fermi level variation due to temperature in extrinsic semiconductors	1	K1	05
2	What do you understand by a resonant cavity? What role does it play in a LASER.	2	K2	05
3	Describe Critical angle with neat labeled diagram. Calculate critical angle for an optical fiber with core refractive index 1.52 and cladding refractive index 1.44. Also find acceptance angle and acceptance cone.	1	K1	05
4	The Hall coefficient of a specimen of doped silicon is found to be 3.66×10^{-4} m^3/C. The resistivity of the specimen is 8.93×10^{-3} Ω-m. Find the mobility and density of the charge carriers, assuming single carrier concentration	2	K2	05

B. Tech. First Year In Semester Examination 1

PHBSC1001: Optics, Semiconductors & Quantum Mechanics

Date: March 27, 2024 **Time: 12.30 to 13.30** **Max. Marks: 15**

Q. N.	**Solve any three.**	CO	B.T. Level	Marks
1	What is antireflection coating? Derive expression for amplitude and phase condition.	1	K1	05
2	Explain Fraunhofer diffraction at a single slit, for formation of central maximum, secondary minima and maxima.	2	K2	05
3	A. Light is incident normally on a grating of total ruled width 5×10^{-3}m with 2500 lines in all. Calculate the angular separation of two sodium lines having wavelengths - λ_1=5890 Å & λ_2=5896 Å in first order spectrum.	3	K3	03
	B. In Newton's rings experiment, radius of curvature of lens is 50 cm, diameter of the 3rd and 23rd ring is 0.181cm and 0.501 cm respectively. Calculate the wavelength of light used?	3	K3	02
4	Describe Heisenberg's uncertainty principle and show that electron cannot exist inside the nucleus.	2	K2	05

Note: CO: Course Outcome; B.T. Level: Bloom's Taxonomy Level

Government College of Engineering, Ch.Sambhajinagar

(An Autonomous Institute of Government of Maharashtra)

B.Tech. First Year (EnTC/CSE/IT)

End semester Examination

Course code- PHBSC1002 **Course title-Optics, Semiconductors Quantum mechanics**

ime: 2.30 Hours 21 MAY 2024 **Max.Marks:60**

"Verify the Course Code and check whether you have got the correct question paper"

N.B:-1. All questions are compulsory

2. Figures to the right indicate full marks

3. Assume suitable data if necessary and state it clearly

4. Use of non-programmable calculator is allowed

Q1	**Attempt any two of the following**	CO	B.T. Level	Mark
a	i) Discuss four level pumping scheme for laser action. ii) Discuss De-Broglie's hypothesis of matter waves.	1	K1	3 3
b	Discuss construction and working of Helium-Neon gas laser, with the help of energy level diagram.	1	K1	6
c	i)With the help of block diagram, discuss working of optical fibre communication system. ii) Describe double refraction phenomenon in birefringent crystal.	1	K1	3 3
Q2	**Attempt any two of the following**			
a	Explain following terms-i)Stimulated emission ii)Population inversion iii)Lasing action	2	K2	6
b	Explain the working of P-N junction on the basis of band structure of p-n junction, when it is forward biased.	2	K2	6
c	i) Derive an expression for resolving power of grating.. ii) Discuss in detail the types of optical fibre, based on refractive index profile.	2	K2	3 3
Q3	**Attempt any two of the following**			
a	i) In Newton's ring experiment the diameters of 10th and 12th dark rings are 0.4cm and 0.7cm respectively. Find the radius of 20th dark ring. ii) Can a thin film of water μ=1.33 formed on a glass window pane μ=1.52 act as an antireflection film? If yes, what should be the minimum thickness of water film? Given λ= 6000Å	3	K3	3 3
b	i) Light is incident normally on grating of total ruled width $5x10^{-3}$m with 2500 lines in all. Calculate the angular separation of two sodium lines in second order spectrum. Given- λ_1=5890Å, λ_2=5896Å	3	K3	3

	ii) Calculate the thickness of a birefringent crystal plate capable of producing a path difference of $\lambda/2$ between e-ray and o-ray propagating through it. Given that μ_o=1.658, μ_e=1.486 λ=5890 Å		
c	i) In N type Germanium sample, the donor density is $10^{21}/m^3$ .It is arranged in a Hall experiment having magnetic field 0.05Tesla and current density 500Amp/m^2 .Find Hall voltage if the sample is 3mm wide. ii) A parallel beam of light λ=5890Å is incident on a thin plate such that the angle of refraction into the plate is 60^0 .Calculate the smallest thickness of film for which the film will appear dark in reflected light.	3	K3
Q4	**Attempt any two of the following**		
a	State and explain Hall effect. Derive an expression for Hall Coefficient R_H.	4	K4
b	Derive an expression for superposition of extraordinary and ordinary waves propagating through a thin plate of birefringent crystal plate. Show how production of elliptically and circularly polarized light takes place.	4	K4
c	Derive an expression for Schrödinger's time dependent wave equation.	4	K4
Q5	**Attempt any two of the following**		
a	i) Classification of solids on the basis of band theory ii) Justify roll of wedge shaped film in Formation of Newton's rings, draw diagram of experimental arrangement.	5	K5
b	Justify the position of Fermi level in intrinsic, p-type, n-type extrinsic semiconductor, with derivation.	5	K5
c	i) Derive the wave equation of electromagnetic wave. ii) Justify transverse nature of electromagnetic wave .	5	K5

Government College of Engineering, Chh. Sambhaji Nagar
(An Autonomous Institute of Government of Maharashtra)
F.Y (CSE/IT) FT (NEP) Examination
ISE - I [CT 1]
MEESC1006: Basics of Mechanical Engineering

ime: 40 Min. *Max. Marks: 10*

"Verify the Course Code and check whether you have got the Correct Question Paper"

N.B.
1. ***Attempt all questions.***
2. ***Figures to the right indicate full marks.***
3. ***Assume suitable data if necessary and state it clearly.***
4. ***Use of Non-Programmable Calculator and Data Sheet is allowed.***

<table>
<tr><th>Q1</th><th>Attempt any TWO</th><th>CO</th><th>B.T. Level</th><th>Marks</th></tr>
<tr><td></td><td>A. State and explain second law of thermodynamics.</td><td>CO1</td><td>BT2</td><td>03</td></tr>
<tr><td></td><td>B. State & Explain Law which governs heat transfer due to convection.</td><td>CO1</td><td>BT1</td><td>03</td></tr>
<tr><td></td><td>C. In an internal combustion engine, during the compression stroke the heat rejected to the cooling water is 50 kJ/kg and the work input is 100 kJ/kg. Calculate the change in internal Energy of the working fluid stating whether it is a gain or loss.</td><td>CO2</td><td>BT5</td><td>03</td></tr>
<tr><th>Q2</th><th>Attempt any ONE</th><th>CO</th><th>B.T. Level</th><th>Marks</th></tr>
<tr><td></td><td>A.. Complete the table for a cyclic process 1-2-3-4
<table>
<tr><th rowspan="2">Process</th><th rowspan="2">Q (KJ)</th><th rowspan="2">W (KJ)</th><th colspan="2">Internal Energy U (KJ)</th></tr>
<tr><th>Initial</th><th>Final</th></tr>
<tr><td>1-2</td><td>0</td><td>-170</td><td>0</td><td>—</td></tr>
<tr><td>2-3</td><td>100</td><td>0</td><td>—</td><td>—</td></tr>
<tr><td>3-4</td><td>0</td><td>———</td><td>—</td><td>70</td></tr>
<tr><td>4-1</td><td>—</td><td>0</td><td>—</td><td>—</td></tr>
</table></td><td>CO2</td><td>BT5</td><td>04</td></tr>
<tr><td></td><td>B. Drive Expression for Overall Heat Transfer coefficient.</td><td>CO2</td><td>BT4</td><td>04</td></tr>
</table>

(Government of Maharashtra)

F.Y (CSE/IT) FT (NEP) Examination

ISE - II [CT 2]

MEESC1006: Basics of Mechanical Engineering

e: 40 Min. *Max. Marks*

"Verify the Course Code and check whether you have got the Correct Question Paper"

N.B.

1. ***Attempt all questions.***
2. ***Figures to the right indicate full marks.***
3. ***Assume suitable data if necessary and state it clearly.***
4. ***Use of Non-Programmable Calculator and Data Sheet is allowed.***

Q1	**Attempt any TWO**	*CO*	*B.T. Level*	*Marks*
	A. What are mountings & accessories in boiler?	*CO3*	*BT1*	*03*
	B. State the function and location of following mountings in boiler 1. Steam Stop Valve 2. Blow off valve 3. Fusible Plug	*CO3*	*BT2*	*03*
	C. Define refrigeration. State its unit. State the COP of refrigeration.	*CO3*	*BT4*	*03*
Q2	**Attempt any ONE**	*CO*	*B.T. Level*	*Marks*
	A. Explain the working of steam power plant in detail.	*CO3*	*BT2*	*04*
	B. Explain the working of Four stroke petrol engine with neat sketch	*CO3*	*BT2*	*04*

(Autonomous Institute of Government of Maharashtra)

End Semester Examination May-June 2024

First Year B. Tech. CSE & IT– NEP – Semester – II

MEESC1006: Basics of Mechanical Engineering

Time: 1.5Hours 15 MAY 2024 *Max. Marks: 30*

"Verify the Course Code and check whether you have got the Correct Question Paper"

N.B.

1. *Attempt all questions.*
2. *Figures to the right indicate full marks.*
3. *Assume suitable data if necessary and state it clearly.*
4. *Use of Non-Programmable Calculator and Data Sheet is allowed.*

Q.1	**Attempt any One**	*CO*	*B.T. Level*	*Marks*
A	Define Absolute pressure, Gauge pressure and vacuum pressure. State different pressure measurement devices and explain Bourdon tube pressure gauge in detail.	*CO1, CO2*	*BT2*	*7.5*
B	Derive the expression for Overall Heat Transfer coefficient and solve the following: A hot plate 1 m × 1.5 m is maintained at 300°C. Air at 20°C blows over the plate. If the convective heat transfer coefficient is 20 W/m^2°C, calculate the rate of heat transfer.	*CO1, CO2*	*BT3*	*7.5*
Q.2	**Attempt any One**	*CO*	*B.T. Level*	*Marks*
A	Define refrigeration. State its unit. State the COP of refrigeration and explain VCRs cycle with neat sketch.	*CO3*	*BT2*	*7.5*
B	Define engine. With neat sketch explain any five parts of Four stroke petrol engine and write down the difference between SI & CI engine.	*CO3*	*BT1*	*7.5*
Q.3	**Attempt any One**	*CO*	*B.T. Level*	*Marks*
A	Explain with neat sketch Slider crank mechanisms, pendulum pump, Oscillating cylinder engine, Whitworth quick return mechanism.	*CO4*	*BT2*	*7.5*
B	Explain different types of couplings with neat diagram.	*CO4*	*BT1*	*7.5*
Q.4	**Attempt any One**	*CO*	*B.T. Level*	*Marks*
A	Explain arc welding in detail also draw and explain different types of welding symbols.	*CO5*	*BT2*	*7.5*
B	Explain with neat sketch any five terminologies used in casting.	*CO5*	*BT1*	*7.5*

Note: CO: Course Outcome; B.T. Level: Bloom's Taxonomy Level

BT1 - Remember, BT2- Understand, BT3 - Apply, BT4 - Analyze, BT5- Evaluate, BT6 - Create

College of Engineering, Chhatrapati Sambhaji nagar
(An Autonomous Institute of Government of Maharashtra)
CSE (NEP)
ISE – II
ETESE 1003: BASICS OF ELECTRONICS ENGINEERING

Time: One Hour Date of Examination: 28/03/2024 *Max. Marks: 15*

"Verify the course code and check whether you have got the correct question paper"

N.B:-

1. *Assume suitable data if necessary and state it clearly*
2. *Use of non-programmable calculator is allowed*
3. *Attempt all questions in sequence*

Qno	Questions	(Cos)	Marks
Q1.	**Attempt any one** a) Represent the decimal numbers 27 and 396 in the following code format i) Binary code ii) BCD code iii) Excess-3 code iv) Gray code v) Octal code vi) hexadecimal code	CO4	[5]
	b) Describe universal gates with truth table and symbol. Why these gates are called as universal? justify	CO4	
Q2.	**Attempt any two** a) Differentiate BJT & FET. (4 points) Explain with diagram CMOS inverter	CO2	[5+5]
	b) Perform addition and subtraction in 2's complement in 8 bit combination. i) 48 - (-23) ii) - 48 - 23 Draw full adder circuit using logic gates write truth table.	CO5	
	c) Demonstrate working of IC 7483 with one example. Write pin function	CO6	

Government College of Engineering, Chhatrapati Sambhaji nagar.

(An Autonomous Institute of Government of Maharashtra)

F.Y B.TECH (CSE/IT) (NEP) Examination

End Semester Examination MAY 2024

ETESC1003: BASICS OF ELCTRONICS ENGINEERING

ime: 2. 30 Hours 17 MAY 2024 *Max. Marks: 60*

"Verify the Course Code and check whether you have got the Correct Question Paper"

B.

Attempt all question.
Figures to the right indicate full marks
Assume suitable data if necessary and state it clearly
Use of Non-Programmable Calculator and Data Sheet is allowed

Q1	**Attempt any TWO**	***CO***	***B.T. Level***	***Marks***
	A) Describe what is P-N junction? Explain the formation of depletion layer in P-N junction..	CO1	K1	***06***
	B) Explain with neat construction and working of LED. Discuss application of LED	CO1	K1	***06***
	C) Differences between three configurations of transistor.	CO2	K2	***06***
Q2	**Attempt any TWO**	***CO***	***B.T. Level***	***Marks***
	A) Explain construction working of n channel depletion type MOSFET. difference between FET & BJT.	CO2	K2	***06***
	B) Describe working principle of common emitter configuration with input output characteristics & derive the expression for amplification factor .	CO3	K2	***06***
	C) Discribe transister as an switch and derive relation between α β and γ.	CO3	K2	***06***
Q3	**Attempt any TWO**	***CO***	***B.T. Level***	***Marks***
	A) Represent the following decimal number into the binary form using i) Binary ii) BCD iii)Excess 3 iv) Gray codes 1] 315 2] 1009 3] 225	CO4	K3	***06***
	B) Justify why NAND and NOR gate is universal gate. Explain with their symbol and truth table.	CO4	K3	***06***
	C) Determine which bit is in error in the even parity. Hamming character is 1100111.	CO4	K3	***06***
Q4	**Attempt any TWO**	***CO***	***B.T. Level***	***Marks***
	A) A bit stream 1101011011 is transmitted using the standard CRC method. The generator polynomial is X^4+X+1. What is actual bit transmitted and determine the data is transmitted with or without error.	CO4	K3	***06***
	B) Perform the subtraction using 2's complement. 1] 32-15 2] -47-35 3] 75-(-56)	CO5	K3	***06***
	C) Illustrate BCD adder with neat and explain with an example.	CO5	K3	***06***

Q5	**Attempt any TWO**	*CO*	*B.T. Level*	*Marks*
	A) Minimize the following expression by using K-map. $F(A,B.C.D) = \sum m\ (0,1,2,3,5,7,8,9,11,14)$	CO5	K2	*06*
	B) Solve the following expression by using Boolean Algebra. 1] $Y= (A+B)(A+\bar{B})(\bar{A}+C)$ 2] $Y= \bar{A}A+\bar{A}\bar{C}+A\bar{B}+\bar{B}\bar{C}+\bar{B}B+\bar{B}C$ 3] $Y= (AB+X+Y+Z)(AB+\bar{X}\bar{Y}\bar{Z})$	CO6	K2	*06*
	C) Justify How to overcome the race around condition in JK flip flop. Explain it with neat diagram and truth table.	CO6	K2	*06*

: CO: Course Outcome; B.T. Level: Bloom's Taxonomy Level